

1. **Rules and selectors:** Writing and linking CSS · Basic selectors: type, class, id, attribute · Combinators: descendant, child and siblings · Pseudo-classes: state and position
2. **Cascade and specificity:** The cascade: which declaration wins · Specificity: the (a, b, c) score · Inheritance: values that flow down the tree
3. **Units and styling:** Units: px, %, em, rem, vw/vh and fr · Colour, text and decoration
4. **Flow, box model and position:** Normal flow and display: block, inline, inline-block, none · The box model: content, padding, border, margin · box-sizing: what width really measures · Margins: centring with auto, and collapsing · Position and overflow: leaving the flow, and content that does not fit
5. **Flexbox:** Flexbox: container, items and the two axes · Flexbox alignment: justify-content, align-items, gap, wrap · Flexible sizes: flex-grow, flex-shrink, flex-basis
6. **Grid:** Grid: tracks, fr and auto-placement · Placing grid items: lines, spans and alignment · Named areas: grid-template-areas · Flexbox or Grid?
7. **Responsive design:** Responsive foundations: viewport tag, fluid widths, fluid images · Media queries and breakpoints

Rules and selectors

Writing and linking CSS

A CSS rule pairs a selector (which elements) with declarations (how they look); rules reach the page inline, in a `<style>` element, or from a .css file linked with `<link rel="stylesheet">`.

- p** The **selector**: which elements the rule styles.
- { ... }** The **declaration block**.
- color: navy;** One **declaration**: a property, a colon, a value and a semicolon.
- /* ... */** A comment. The HTML form `<!-- -->` and the JavaScript form `//` do not work in CSS.

Three ways to attach CSS

Where	How	Reaches
Inline	<code>style="color: red"</code> on one element	That element only; it beats every selector, so avoid it outside quick tests
Internal	A <code><style></code> element in the <code><head></code>	One page
External	<code><link rel="stylesheet" href="styles.css"></code> in the <code><head></code>	Every page that links it: the standard way

```

<!-- index.html, fixtures.html, login.html: the same line in every <head> -->
<head>
  <meta charset="UTF-8">
  <title>Riverside FC</title>
  <link rel="stylesheet" href="styles.css">
</head>

/* styles.css: one file, every page */
nav a {
  color: #1A1F6C;          /* text colour */
  text-decoration: none; /* no underline */
}
```

Example. A club site has `index.html`, `fixtures.html` and `login.html`, each with the same `<link rel="stylesheet" href="styles.css">`. Changing `nav a { color: ... }` once in `styles.css` recolours the navigation on all three pages, and the browser reuses its cached copy of the file on the next page.

Common mistake. Writing `<style src="styles.css">` or putting the `<link>` in `<body>`. `<style>` has no `src` attribute: external CSS only loads through `<link rel="stylesheet">` in the `<head>`. Check the path too: `href="css/styles.css"` is relative to the HTML file, and a wrong path fails silently (the Network tab in DevTools shows a 404).

Basic selectors: type, class, id, attribute

A selector picks the elements a rule applies to: by tag (p), class (.card), id (#hero), attribute (a[target]) or all of them (*).

The basic selectors

Selector	Matches	Example
Type	every element with that tag	p, nav, button
Class	every element whose class list contains the name	.card, .btn-primary
Id	the one element with that id	#hero
Attribute	elements that have an attribute, or a given value	a[target], input[type="email"]
Universal	every element	*
Compound	one element meeting all the parts	a.btn, .card.featured
Group (comma)	elements matching any of the selectors	h1, h2, h3

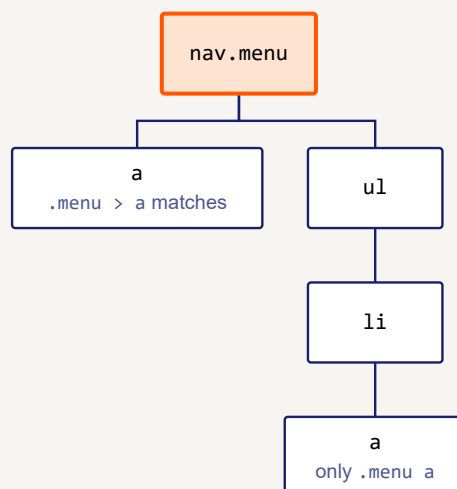
- **Classes** are the workhorse: one element can carry several (`class="card featured"`) and a class can be reused on any number of elements.
- **Ids** are unique in the page and score high in specificity, so styles hung on them are hard to override. Keep ids for link targets and scripts (see [Attributes](#)).
- **No space compounds:** `article.card` is an `<article>` that also has class `card`; `.card.featured` is one element with both classes.
- **A comma groups** independent selectors that share declarations: `h1, h2, h3 { ... }`.

Example. For `<a href="/tickets" target="_blank" class="btn primary">Buy</a>`, all of these match: `a`, `.btn`, `.primary`, `a.btn`, `.btn.primary`, `a[target]`, `a[target="_blank"]`. The selector `.btn .primary` does **not** match it: the space asks for a `.primary` element **inside** a `.btn`.

Common mistake. Confusing `.card.featured` (no space: one element with both classes) with `.card .featured` (space: a `.featured` element somewhere inside a `.card`). One space changes the meaning completely, and the wrong one simply matches nothing, with no error.

Combinators: descendant, child and siblings

A combinator joins two selectors by how the elements are related in the HTML tree: anywhere inside (a space), direct child (`>`), next sibling (`+`) or any later sibling (`~`).



A space reaches any depth; `>` reaches only one level down.

The four combinators

Combinator	Written	Matches the B that is...
Descendant	A B (space)	anywhere inside an A, at any depth
Child	A > B	a direct child of an A (one level down)
Next sibling	A + B	immediately after an A, with the same parent
Later sibling	A ~ B	after an A (not necessarily right after), with the same parent

- **Read it right to left:** in `nav a`, the rightmost part (`a`) is the element being styled; everything to its left is context.
- **No specificity:** combinators add nothing to the score; only the selectors around them count (see [Specificity](#)).

Example. `.card + .card { margin-top: 1rem; }` puts space **between** consecutive cards but not above the first one: the first card has no `.card` right before it, so it is not matched. `nav > ul` styles the menu list directly inside `<nav>` but not a sub-menu `<ul>` nested deeper.

Common mistake. Writing `.menu > a` for `<nav class="menu"><ul><li><a>...</a></li></ul></nav>`. The link is a great-grandchild of `.menu`, not a direct child, so nothing matches. Use the descendant combinator `.menu a`, or spell out the whole path `.menu > ul > li > a`.

Pseudo-classes: state and position

A pseudo-class such as `:hover`, `:focus`, `:first-child` or `:nth-child(odd)` matches an element by its current state or its position among its siblings: information that is not written in the HTML.

Pseudo-classes you will use constantly

Pseudo-class	Matches
<code>:hover</code>	while the mouse pointer is over the element
<code>:focus</code>	while the element has keyboard focus (a clicked or tabbed-to input, button or link)
<code>:focus-visible</code>	focus that the browser judges should be visible (typically keyboard navigation, not mouse clicks)
<code>:active</code>	while the element is being pressed
<code>:visited</code>	a link to a page already visited
<code>:first-child</code> / <code>:last-child</code>	the first / last child of its parent
<code>:nth-child(odd)</code> , <code>:nth-child(3n)</code>	children by position: odd ones, every third...
<code>:not(.done)</code>	elements that do not match the selector inside

- **One colon, no space:** `button:hover`, `input:focus`, `li:first-child`.
- **Pseudo-elements** use two colons and style a *part* of an element instead of a state: `p::first-line`, or `::before` and `::after`, which insert generated content.
- **Link states in this order:** `:link`, `:visited`, `:hover`, `:active`. They have the same specificity, so the later one wins when two apply at once, such as a visited link that is hovered (see [Specificity](#)).

Example. A sign-in form: `input:focus { border-color: #FF5700; outline: 3px solid #FFD2B8; }` shows clearly which field is being typed in; `button:hover { background: #1A1F6C; }` gives visual feedback; `tr:nth-child(even) { background: #F4F4F8; }` stripes a fixtures table so rows are easy to follow.

Common mistake. Removing the focus ring with `outline: none` and putting nothing in its place: keyboard users can no longer see where they are. If the default ring clashes with your design, **replace** it (`:focus-visible { outline: 3px solid #FF5700; }`). Also watch the space: `button :hover` (with a space) means "a hovered element **inside** a button", not a hovered button.

Cascade and specificity

The cascade: which declaration wins

When several declarations set the **same property** on the **same element**, the cascade chooses one by asking four questions in order: **origin** and **!important**, **inline style**, **specificity**, and **source order**.

The four questions, in order

- 1. Origin and importance:** your styles beat the browser's defaults; any **!important** declaration beats every normal one.
- 2. Inline style:** a `style=""` attribute beats any selector.
- 3. Specificity:** the selector with the higher (a, b, c) score wins (see [Specificity](#)).
- 4. Source order:** the declaration that appears last wins.

Example. With `.btn { background: grey; }` followed by `.primary { background: orange; }`, the element `<a class="primary btn">` is **orange**: both selectors score (0,1,0), so the later rule wins. Swap the two rules in the file and it turns grey; swapping the class names in the HTML changes nothing.

Common mistake. Fixing a conflict with **!important**. It wins today, but the next time you need an override you need another **!important**, and soon nothing can be changed without one. Look at the Styles pane in DevTools instead: the losing declaration is struck through, so you can see which rule beat it and fix the selector or the order.

Specificity: the (a, b, c) score

Specificity is a three-number score (**a, b, c**) for every selector, compared column by column from the left; the higher score wins.

- a** Id selectors: `#hero`.
- b** Class selectors (`.card`), attribute selectors (`[type="email"]`) and pseudo-classes (`:hover`).
- c** Type selectors (`p`, `li`) and pseudo-elements (`::before`).
- Nothing** The universal selector `*` and the combinators (space, `>`, `+`, `~`).

Worked scores

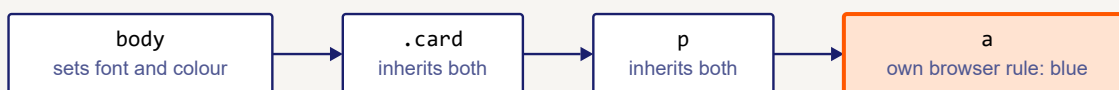
Selector	(a, b, c)	Why
<code>*</code>	(0,0,0)	universal counts nothing
<code>p</code>	(0,0,1)	one type
<code>nav ul li a</code>	(0,0,4)	four types; the spaces count nothing
<code>.intro</code>	(0,1,0)	one class
<code>li:first-child</code>	(0,1,1)	a pseudo-class counts like a class
<code>a[target]</code>	(0,1,1)	an attribute selector counts like a class
<code>.card > p</code>	(0,1,1)	the <code>></code> counts nothing
<code>.nav .link:hover</code>	(0,3,0)	two classes plus one pseudo-class
<code>#hero p</code>	(1,0,1)	one id, one type
<code>#hero .intro</code>	(1,1,0)	beats <code>#hero p</code> : equal a, higher b

Example. Three rules colour the same link: `nav a { color: black; }` scores (0,0,2); `.link { color: navy; }` scores (0,1,0); `#main-nav a { color: orange; }` scores (1,0,1). The link is **orange**, wherever the rules appear in the file. Remove the id rule and it is **navy**: one class beats any number of type selectors.

Common mistake. Treating the score as one decimal number ("`#hero p` is worth 101 points, so eleven classes, worth 110, beat it"). The columns are compared separately and never carry: one id beats any number of classes. The practical lesson is to keep selectors **short and flat** (mostly single classes) so that any rule can be overridden by another single class placed later.

Inheritance: values that flow down the tree

Text properties such as color, font-family, font-size and line-height **are inherited**: an element with no value of its own takes its parent's. **Box properties** such as margin, border and background **are not**.



Values flow down until an element has a rule of its own.

Inherited color, font-family, font-size, font-weight, line-height, text-align, list-style, cursor, visibility.

Not inherited margin, padding, border, width, height, background, display, box-shadow, border-radius.

- **The weakest value:** any rule that targets the element itself, even `p { }` or a browser default, beats an inherited value. That is why links stay blue inside a red paragraph.
- **Form controls** get their own font from the browser: `write button, input, select, textarea { font: inherit; }` to make them follow the page.
- **Keywords:** `inherit` takes the parent's value, even for a non-inherited property; `initial` the property's starting value; `unset` inherits if the property normally inherits, else `initial`.

Example. In `<article class="card"><p>Text with <a href="#">a link</a></p></article>`, the rule `.card { color: darkred; border: 1px solid; }` makes the paragraph text dark red (inherited) but draws **one** border, around the card only (not inherited). The link keeps the browser's blue until you write `.card a { color: inherit; }`.

Common mistake. Expecting `.card { color: red; }` to colour the links inside the card, or the login button to use the page font. Both have browser defaults that target them directly, and a direct rule always beats an inherited value. Target them yourself, or use `inherit`.

Units and styling

Units: px, %, em, rem, vw/vh and fr

Lengths are either absolute (px) or relative to something: the containing block (%), a font size (em, rem), the viewport (vw, vh) or, in grid tracks only, the free space (fr).

Assuming the browser default root font size of 16px

Unit	Relative to	Example	Typical use
px	nothing (a CSS pixel)	<code>border: 1px solid</code>	borders, shadows, small fixed details
%	the containing block	<code>max-width: 100%</code>	fluid widths, images
em	the element's own font size	<code>padding: 0.5em 1em</code>	spacing that should scale with that element's text
rem	the root (<code><html></code>) font size	<code>font-size: 1.25rem = 20px</code>	font sizes, consistent spacing
vw / vh	1% of the viewport width / height	<code>height: 100vh</code>	full-screen sections
fr	a share of the free space in a grid container	<code>grid-template-columns: 2fr 1fr</code>	grid tracks only

- **rem or em:** `rem` refers to the font size of `<html>` (16px by default); `em` to the element's own font size, or to the parent's inside font-size, which is why nested `em` font sizes compound.
- **Percentages:** a % width is a share of the parent's content width; a % padding or margin is a share of the parent's **width**, even at the top and bottom.
- **Zero needs no unit:** `margin: 0`.

Example. With the default 16px root: `h1 { font-size: 2rem; }` is 32px. A button with `font-size: 0.875rem; padding: 0.5em 1em;` has 14px text and padding of 7px (top/bottom) by 14px (left/right): the `em` padding is computed from the button's own 14px font size, so a bigger button gets proportionally bigger padding automatically.

Common mistake. Nesting `em` font sizes: with `li { font-size: 0.8em; }`, a list inside a list item is $0.8 \times 0.8 = 0.64$ of the original size, and a third level 0.51. `em` compounds through the tree; `rem` always goes back to the root, which is why `rem` is the safer default for font sizes. Also, `fr` is not a general length: `width: 1fr` is invalid and ignored.

Colour, text and decoration

The everyday styling properties: colours, fonts, text, lists and decoration such as rounded corners and shadows.

Properties and typical values

Property	What it does	Example
<code>color</code>	text colour (inherited)	<code>color: #222;</code>
<code>background-color</code>	fills content and padding	<code>background-color: #F4F4F8;</code>
<code>font-family</code>	font stack (inherited)	<code>font-family: Arial, sans-serif;</code>
<code>font-size / font-weight</code>	size / boldness	<code>font-size: 1.125rem; font-weight: 700;</code>
<code>line-height</code>	space between lines; unitless = $\times$ font size	<code>line-height: 1.5;</code>
<code>text-align</code>	aligns inline content inside the box	<code>text-align: center;</code>
<code>text-decoration</code>	underline (or not) on text, e.g. links	<code>text-decoration: none;</code>
<code>list-style</code>	list markers (bullets, numbers)	<code>list-style: none;</code>
<code>border-radius</code>	rounds the corners	<code>border-radius: 8px;</code>
<code>box-shadow</code>	x-offset y-offset blur colour	<code>box-shadow: 0 2px 8px rgb(0 0 0 / 0.15);</code>
<code>cursor</code>	mouse pointer shape over the element	<code>cursor: pointer;</code>

- **Colours:** a name (navy), hexadecimal `#RRGGBB` with two digits each for red, green and blue (`#1A1F6C`), or `rgb(26 31 108)`. A fourth value adds transparency: `rgb(0 0 0 / 0.15)` is black at 15%, ideal for shadows. Keep enough contrast between text and background.
- **Font stacks** list fallbacks: in `font-family: "Segoe UI", Arial, sans-serif;` the browser uses the first font installed on the device. End with a generic family (`serif`, `sans-serif`, `monospace`) and quote names with spaces.

Example. A nav without bullets: `nav ul { list-style: none; margin: 0; padding: 0; }` and `nav a { text-decoration: none; color: white; }`. A card: `.card { background: white; border-radius: 8px; box-shadow: 0 2px 8px rgb(0 0 0 / 0.15); padding: 1rem; }`

Common mistake. Writing `text-decoration: none` on a list to remove the bullets: that property removes the **underline** of links. The bullets belong to the list, so it is `list-style: none` on the `<ul>`, usually together with `padding: 0; margin: 0;` to remove the indentation the browser added to make room for them.

Flow, box model and position

Normal flow and display: block, inline, inline-block, none

`display` **sets how a box takes part in layout:** **block** boxes stack and fill the width, **inline** boxes flow inside lines of text, **inline-block** flows in a line but accepts a size, and **none** removes the element.

Block by default `<p>`, `<div>`, `<section>`, `<ul>`, `<form>`, `<h1>`...

Inline-level by default `<a>`, `<span>`, `<strong>`, `<label>`, `<img>`, `<input>`, `<button>`...

How each value behaves

Value	Starts a new line?	Default width	width / height	Vertical margin
<code>block</code>	yes	fills the container	applied	applied (and may collapse)

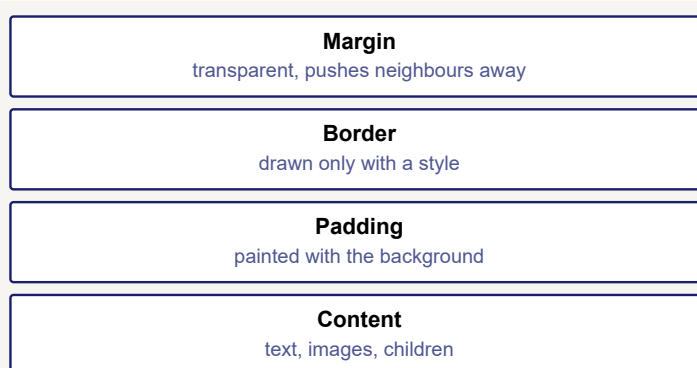
Value	Starts a new line?	Default width	width / height	Vertical margin
inline	no	its content	ignored	ignored (padding paints but does not push lines apart)
inline-block	no	its content	applied	applied
none	—	removed from layout and from screen readers	—	—

Example. A form where each `<label>` sits on its own line above its field: `label { display: block; margin-bottom: 4px; }` and `input { display: block; width: 100%; }`. Nav links that need padding and a clickable area larger than the text: `nav a { display: inline-block; padding: 8px 12px; }`.

Common mistake. Setting width, height or margin-top on an `<a>` or a `<span>` and seeing nothing happen: inline boxes size to their text and ignore those properties. Switch to `inline-block` or `block` (or lay out the parent with Flexbox). Also, `visibility: hidden` is not the same as `display: none`: hidden elements are invisible but **keep their space**.

The box model: content, padding, border, margin

Every element is drawn as a rectangle of four layers, inside out: **content, padding, border and margin**.



From the outside in: margin, border, padding, content.

- padding: 10px** All four sides.
- padding: 10px 20px** Top and bottom 10px, left and right 20px.
- padding: 10px 20px 5px** Top 10px, left and right 20px, bottom 5px.
- padding: 10px 20px 5px 0** Top, right, bottom, left: clockwise from the top. margin takes the same 1 to 4 values.

- `border: 2px solid #ccc` sets width, style and colour; without a style (solid, dashed, dotted...) no border is drawn at all.
- To see the layers** of any element, select it in DevTools and open the **Computed** tab: its box diagram shows the exact sizes.

Example. `.card { padding: 16px; border: 1px solid #ddd; margin-bottom: 12px; background: white; }` gives each card 16px of white breathing room around its text, a thin grey frame, and 12px of empty space before the card that follows it.

Common mistake. Using padding to separate two cards. Padding is **inside** the card: each card grows and its background grows with it, but the cards still touch. Space **between** boxes is margin, or gap when the parent is a flex or grid container.

box-sizing: what width really measures

box-sizing **decides which layers** width **and** height **measure**: content-box (the default) sizes only the content, so

padding and border are added on top; border-box makes padding and border fit inside the declared size.

width: 300px; padding: 20px; border: 5px solid;

	content-box (default)	border-box
Content width	300px	$300 - 2 \times 20 - 2 \times 5 = 250\text{px}$
Visible width (border to border)	$300 + 2 \times 20 + 2 \times 5 = 350\text{px}$	300px
Margin	added outside	added outside

```
*,
*::before,
*::after {
  box-sizing: border-box;
}
```

Example. Two columns with width: 50%; padding: 20px; side by side. With content-box each one is 50% + 40px wide, the pair is wider than the container and the second column wraps below the first. With border-box each is exactly 50% and they fit.

Common mistake. Thinking border-box also includes the margin. It does not: width: 100%; margin: 0 16px; still overflows its parent by 32px. A block box already fills the available width on its own, so the fix is to remove the width (or use max-width).

Margins: centring with auto, and collapsing

margin: 0 auto **centres a block that is narrower than its container (it needs a width or max-width)**; **vertical margins between stacked blocks collapse, so the gap is the larger margin, not the sum.**

Collapsing margins

- **Stacked blocks:** when the bottom margin of one block meets the top margin of the next, only the larger is kept. h2 { margin-bottom: 24px; } above p { margin-top: 16px; } leaves 24px, not 40px.
- **Through a parent:** a child's top margin can collapse through a parent with no border, padding or content above it, so the space appears outside the parent. Padding on the parent, or making it a flex or grid container, stops it.
- **Never** horizontally, and never between the items of a flex or grid container.

Example. A sign-in card: .login { max-width: 420px; margin: 2rem auto; padding: 2rem; border: 1px solid #ddd; }. On a wide screen it is 420px wide with equal space on both sides; on a 360px phone it simply takes the full width, because max-width is a ceiling, not a fixed size.

Common mistake. Using margin: 0 auto on a box without a width (a block already fills its container, so there is no leftover space to share) or on an inline element such as <a> or (horizontal auto margins do nothing on inline boxes). Give the box a max-width, or centre it from the parent with Flexbox.

Position and overflow: leaving the flow, and content that does not fit

position **lets a box leave normal flow or shift from its place**, and overflow **decides what happens to content that does not fit its box.**

The values of position

Value	Keeps its space in the flow?	Offsets measured from	Typical use
static (default)	Yes	Offsets are ignored	Normal flow
relative	Yes, its original place	Its own normal position	Small nudges; a reference box for absolute children
absolute	No	The nearest positioned ancestor (any value but static), else the page	A badge or a close button on a card corner
fixed	No	The viewport: it stays put while the page scrolls	A back-to-top button, a cookie bar

Value	Keeps its space in the flow?	Offsets measured from	Typical use
sticky	Yes	Scrolls normally, then sticks at its offset inside its container	A table header, a section title bar

- **The usual pair:** `position: relative` on the card, with no offsets so it stays in place, and `position: absolute; top: 8px; right: 8px;` on the badge inside it.
- **z-index** orders overlapping boxes: a higher value is drawn in front. It works on positioned elements and on flex and grid items.

Content that does not fit: overflow

visible (default) The content spills out of the box and overlaps whatever follows.

hidden The content is cut off at the box edge.

auto A scrollbar appears only when the content is too big.

scroll Scrollbars always show.

Example. A product card with a "Sale" badge: `.card { position: relative; }` and `.badge { position: absolute; top: 8px; right: 8px; }`. The badge sits on the card's corner whatever the card's size, and the card's text flows as if the badge were not there. In the Try it box, remove `position: relative` from `.card` and the badge jumps to the corner of the preview.

Common mistake. Giving the badge `position: absolute` but forgetting `position: relative` on the card. With no positioned ancestor the offsets are measured from the page, so every badge piles up in the page's top-right corner. A similar trap: `position: sticky` inside a parent with `overflow: hidden` sticks to that parent, which never scrolls, so it seems to do nothing.

Flexbox

Flexbox: container, items and the two axes

`display: flex` **makes an element a flex container** and its direct children **flex items**, laid out along a **main axis** chosen with `flex-direction`; the **cross axis** is perpendicular to it.

What each `flex-direction` does to the axes

flex-direction	Main axis	Cross axis	<code>justify-content</code> moves items	<code>align-items</code> moves items
row (default)	left → right	top → bottom	horizontally	vertically
row-reverse	right → left	top → bottom	horizontally	vertically
column	top → bottom	left → right	vertically	horizontally
column-reverse	bottom → top	left → right	vertically	horizontally

- **Direct children only:** in `<nav><ul><li>`, `display: flex` on `<nav>` makes the `<ul>` the only flex item, and the `<li>`s stay stacked. Make the `<ul>` the container to put the links in a row.
- `row-reverse` **and** `column-reverse` flip the visual order only. Screen readers and the Tab key still follow the HTML order, so never use them to fix content that is in the wrong order.

Example. A site header: `header { display: flex; justify-content: space-between; align-items: center; }` puts the `<h1>` on the left and the `<nav>` on the right (main axis), both vertically centred (cross axis). Inside it, `nav ul { display: flex; gap: 1rem; list-style: none; }` turns the links into a row.

Common mistake. Believing `justify-content` is always horizontal. It works along the **main axis**: after `flex-direction: column` it moves items vertically and `align-items` moves them horizontally. And vertical centring in a column only shows if the container is taller than its content (give it a height or `min-height`).

Flexbox alignment: justify-content, align-items, gap, wrap

`justify-content` distributes the items along the main axis, `align-items` positions them on the cross axis, `gap` sets the space between them, and `flex-wrap: wrap` lets them flow onto new lines.

`justify-content` values (main axis)

Value	Leftover space goes...
<code>flex-start</code> (default)	all at the end; items packed at the start
<code>center</code>	half before the first item, half after the last
<code>flex-end</code>	all at the start; items packed at the end
<code>space-between</code>	between items only; first and last touch the edges
<code>space-around</code>	equal space on both sides of every item, so edges get half a gap
<code>space-evenly</code>	equal gaps everywhere, edges included

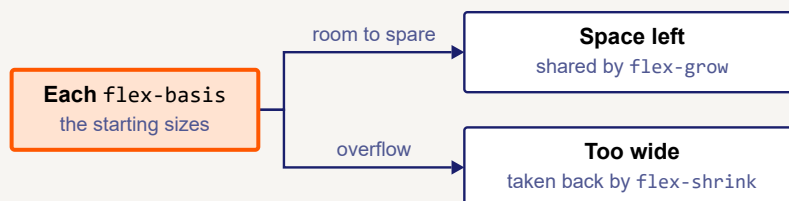
- `align-items: stretch`, the default, makes items without a set height as tall as the line: that is why cards in a flex row come out equal-height for free. `align-self` overrides it for one item.
- **No wrapping by default:** with `flex-wrap: nowrap` items shrink to fit and may overflow; `flex-wrap: wrap` starts a new line instead. With several lines, `align-content` positions the lines.
- `gap` on the container puts space only between items, in both directions.

Example. A tag cloud of `#javascript`, `#css`, `#webdev`, `#accessibility`, `#performance`: `.tags { display: flex; flex-wrap: wrap; gap: 0.5rem; }` keeps each tag at its natural width and moves to a new line whenever the next tag no longer fits, with the same 0.5rem gap horizontally and vertically.

Common mistake. Spacing items with `margin-right` on each one: the last item gets a useless margin, and wrapped lines get no vertical space. `gap` on the container puts space **only between** items, in both directions. Another one: adding `align-items: center` to a row of cards "to tidy it up" and losing their equal heights, which came from the default stretch.

Flexible sizes: flex-grow, flex-shrink, flex-basis

Each flex item starts from a size (`flex-basis`), takes a share of any leftover space (`flex-grow`) and gives back a share of any overflow (`flex-shrink`); the shorthand `flex` sets all three.



Start from the basis, then share out the difference.

Common `flex` shorthands

Shorthand	Expands to	Meaning
(default)	<code>0 1 auto</code>	content size, may shrink, never grows
<code>flex: 1</code>	<code>1 1 0%</code>	equal shares of all the space
<code>flex: 2</code>	<code>2 1 0%</code>	twice the share of a <code>flex: 1</code> sibling
<code>flex: auto</code>	<code>1 1 auto</code>	content size plus a share of the leftover
<code>flex: none</code>	<code>0 0 auto</code>	exactly its content size; never grows or shrinks

Example. A search bar: `.search { display: flex; gap: 8px; }, .search input { flex: 1; }, .search button { flex: none; }`. The button keeps its natural width and the input takes **all** the rest, on any screen width. In a 600px container with no gap, two items with `flex: 1` and `flex: 2` become 200px and 400px.

Common mistake. Expecting flex: 1 items to be equal when one contains a long unbreakable word, a URL or a wide image. A flex item will not shrink below its content's minimum width (min-width: auto by default), so it pushes its siblings narrower or overflows. Add min-width: 0 (and, for text, overflow-wrap: anywhere) to the item.

Grid

Grid: tracks, fr and auto-placement

display: grid turns an element's children into **grid items** placed in cells formed by column and row **tracks**, sized with grid-template-columns / grid-template-rows in px, %, auto or fr (a share of the free space).



fr is computed last, from whatever space is left.

repeat(3, 1fr)

Short for 1fr 1fr 1fr.

minmax(200px, 1fr)

A track never narrower than 200px, growing up to 1fr.

repeat(auto-fit, minmax(220px, 1fr))

As many 220px-or-wider columns as fit: a card gallery goes from 4 columns to 1 without any media query.

gap

Space between tracks (or row-gap / column-gap), never at the outer edges.

Example. The Newsroom <main> with display: grid; grid-template-columns: 2fr 1fr; gap: 2rem; on a 930px container: 930 – 32 = 898px free, so the article column is 598.7px and the sidebar 299.3px, and both keep that 2:1 ratio at any width.

Common mistake. Putting grid-template-columns: 2fr 1fr on a <main> that has **three** children and expecting "articles left, sidebar right". Auto-placement fills the cells in order: the first child takes the left cell, the second the right cell, and the third wraps to the left cell of row 2. Make the extra child span both columns or place the items explicitly (see [Placing grid items](#)).

Placing grid items: lines, spans and alignment

Grid lines are numbered from 1 at the start edge (and from –1 at the end edge). grid-column / grid-row place an item between two lines (1 / 3) or make it span several tracks; justify-items / align-items position the content inside each cell.

Placement in a 3-column grid

Declaration	Covers
grid-column: 1 / 2	column 1 only (line 1 to line 2)
grid-column: 1 / 3	columns 1 and 2
grid-column: 1 / -1	all three columns (first line to last line)
grid-column: span 2	two columns, starting wherever auto-placement puts it
grid-column: 2 / span 2	columns 2 and 3
grid-row: span 2	two rows tall

- span n covers n tracks from wherever the item would start: grid-column: span 2. The forms mix: 2 / span 2 starts at line 2 and covers two columns. grid-row works the same way vertically.
- Inside its cell** an item is positioned by justify-items (horizontally) and align-items (vertically), set on the container and both stretch by default. justify-self and align-self override them for one item; justify-items has no effect in a flex container.

Example. A gallery of six cards in repeat(3, 1fr): give the first card grid-column: span 2; grid-row: span 2; and it becomes a featured tile covering a 2×2 block, while the remaining cards auto-place around it with no extra HTML.

Common mistake. Counting columns instead of lines: `grid-column: 1 / 2` covers just **one** column. To cover columns 1 and 2 write `1 / 3` (or `1 / span 2`). Another one: using `1 / -1` with only implicit columns: `-1` refers to the last line of the **explicit** grid (the one defined by `grid-template-columns`).

Named areas: grid-template-areas

`grid-template-areas` **draws the layout as text, one quoted string per row and one name per cell; each item is then placed with** `grid-area: name`.

- **The same number of cells** in every string, matching the columns.
- **Rectangles only:** each named area must form a rectangle, never an L shape. A . marks an empty cell; a name repeated across neighbouring cells spans them.
- **A broken template** makes the whole declaration invalid, so it is ignored.
- **Names without quotes** on the items: `.site-header { grid-area: header; }`. Every child needs one: a child without `grid-area` is auto-placed into a free cell.

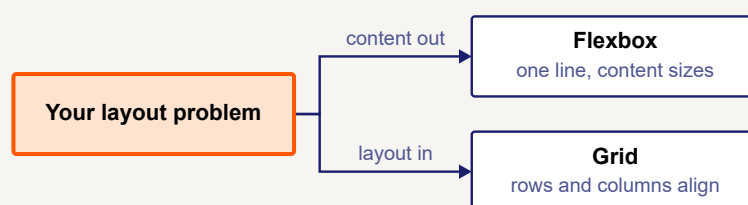
```
.page {
  display: grid;
  grid-template-columns: 240px 1fr;
  grid-template-areas:
    "header header"
    "sidebar content"
    "footer footer";
  gap: 1.5rem;
}
.site-header { grid-area: header; }
.sidebar      { grid-area: sidebar; }
.content      { grid-area: content; }
.site-footer  { grid-area: footer; }
```

Example. In this layout the header and the footer span both columns because their names repeat across their row, and the sidebar and the content share the middle row. Redrawing the sketch as one column ("`header`" "`content`" "`sidebar`" "`footer`", with `grid-template-columns: 1fr`) moves the sidebar below the content without touching the HTML.

Common mistake. Defining the areas on the container but forgetting `grid-area` on one item (typically the header or footer). That item is auto-placed into the first free cell, often squeezed into the sidebar column, and the named area stays empty. Every region in the sketch needs a matching `grid-area` on exactly one element.

Flexbox or Grid?

Use **Flexbox** when items flow along one line and their content should decide their size; use **Grid** when rows and columns must line up and the container should decide the sizes. Most pages use both: Grid for the page skeleton, Flexbox inside the regions.



Choose by the shape of the problem.

Typical choices

Situation	Tool	Why
Logo on the left, nav on the right	Flexbox	one row; <code>justify-content: space-between</code>
Links in a nav bar, buttons in a toolbar	Flexbox	one row of content-sized items with gap
Tags or chips of different lengths	Flexbox + <code>flex-wrap</code>	each item keeps its natural width
Input stretching next to a button	Flexbox	<code>flex: 1</code> on the input

Situation	Tool	Why
Page skeleton: header / sidebar / main / footer	Grid	two dimensions, named areas
Card gallery whose columns line up	Grid	columns shared by every row
Content + sidebar with a fixed ratio	Grid	grid-template-columns: 2fr 1fr

Example. A club homepage: body (or a wrapper) is a grid with areas header / main / footer; the header is a flex row (crest left, nav right); the fixtures list on the main area is a grid of match cards; each match card is a flex row (home team, score, away team) with `justify-content: space-between`.

Common mistake. Believing one replaced the other ("Grid is the new Flexbox" or "just use Flexbox for everything"). Building a card gallery with wrapped Flexbox gives a last row whose cards stretch wider than the rest; building a nav bar with Grid forces you to count columns every time a link is added. Choose by the shape of the problem.

Responsive design

Responsive foundations: viewport tag, fluid widths, fluid images

A responsive page adapts to any screen width. It needs the **viewport** `<meta>` tag, widths that are relative instead of fixed (`%`, `fr`, `max-width`), and images capped with `img { max-width: 100%; height: auto; }`.

The viewport tag `<meta name="viewport" content="width=device-width, initial-scale=1">` in the `<head>` of every page: "this page is designed for your real width".

Fluid widths Blocks fill their container, grids share space with `fr`, and a content wrapper is capped with `max-width` and centred with `margin: 0 auto`.

Fluid images `max-width: 100%` lets an image shrink with its container but never grow beyond its natural size; `height: auto` recalculates the height so the proportions are kept.

```
<!-- in the <head> of EVERY page -->
<meta name="viewport" content="width=device-width, initial-scale=1">

/* styles.css */
img {
  max-width: 100%; /* shrink with the container, never stretch */
  height: auto; /* keep the aspect ratio */
}
.container {
  max-width: 1100px; /* a ceiling, not a fixed width */
  margin: 0 auto;
  padding: 0 1rem;
}
```

Example. An `` in a 360px-wide phone column: with `max-width: 100%; height: auto`; it is drawn 360 × 180, keeping its 2:1 shape. To test, open DevTools and toggle the device toolbar (`Ctrl+Shift+M` / `Cmd+Shift+M`) to emulate a phone; without the viewport tag, the emulated phone shows the zoomed-out desktop layout.

Common mistake. Writing `max-width: 100%` but forgetting `height: auto` on an image that has width and height attributes: the width shrinks but the height stays at the attribute value, so the picture is squashed. And using `width: 100%` instead of `max-width: 100%`: small images such as logos and icons are stretched up and blurred.

Media queries and breakpoints

A media query such as `@media (max-width: 768px) { ... }` wraps rules that apply only while a condition about the

device is true, most often the viewport width. The width at which the layout changes is called a **breakpoint**.



Mobile-first: a narrow base, enhanced as the screen gets wider.

max-width: 768px A viewport 768px wide or less. **Desktop-first** stylesheets write the wide layout as the base and override it with these.

min-width: 600px 600px or wider. **Mobile-first** stylesheets write the one-column layout as the base and add columns with these; the narrow layout is the simplest, so there is less to override.

- **No extra specificity:** a rule inside a query competes like any other, so the `@media` block must come **after** the base rules it overrides.
- **More conditions:** and combines them, as in `(min-width: 600px)` and `(max-width: 900px)`; `@media print` and `(prefers-color-scheme: dark)` test more than width.
- **Named areas:** a new `grid-template-areas` sketch inside a query moves whole regions at once (see [Named areas](#)).
- **Breakpoints** belong where *your* layout starts to look wrong, not at specific phone models.

```

/* Desktop-first: wide layout is the base */
main { display: grid; grid-template-columns: 2fr 1fr; gap: 2rem; }

@media (max-width: 768px) {           /* 768px or narrower */
  main { grid-template-columns: 1fr; }
}

/* Mobile-first: one column is the base */
.cards { display: grid; gap: 1rem; }

@media (min-width: 600px) {           /* 600px or wider */
  .cards { grid-template-columns: repeat(2, 1fr); }
}
@media (min-width: 1000px) {
  .cards { grid-template-columns: repeat(3, 1fr); }
}
  
```

Example. A typical news-style page: main is a two-column grid (2fr 1fr); at 768px or less, `@media (max-width: 768px) { main { grid-template-columns: 1fr; } }` turns it into one column, so the `<aside>` drops below the articles. Resize the window (or use the DevTools device toolbar) to check that the switch really happens.

Common mistake. Putting the `@media` block at the **top** of the stylesheet. The base rule `main { grid-template-columns: 2fr 1fr; }` further down has the same specificity and comes later, so it wins at every width and the breakpoint seems broken. Media queries go after the rules they modify. The other classic cause is a missing viewport `<meta>` tag (see [Responsive foundations](#)).